Entropy-Weighted Multi-Order Markov Chains with Dual-Memory for Mobility Prediction

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Abstract

Mobility prediction methods generally follow two paradigms: individual models, which capture user-specific behaviour but suffer from cold-start, and collective models, which exploit population-level patterns but often lack online adaptability. We propose a Dual-Memory Markov Chain (DM-MC), a model that combines a population-based Long-Term Memory with an online Short-Term Memory built from the current trajectory. Multi-order Markov predictions are fused through entropy-based weighting, allowing adaptive selection of both memory source and Markov order. Experiments demonstrate competitive performance on next-node and auto-regressive prediction tasks. DM-MC offers a simple and interpretable dual-memory approach to mobility modelling.

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1 Introduction

Understanding and anticipating human movement has emerged as a critical problem covering a wide range of real-world applications, from intelligent transportation systems and traffic management to personalised navigation and urban planning, making the accurate prediction of the next node/position in a road network trajectory a cornerstone of modern mobility modelling. First-order Markov chains suffer from the memoryless assumption, while higher and variable-order extensions (e.g. AKOM [10], CPT+ [4], MOGen [2], and de Bruijn graphs [12, 17]) improve sequential modelling but remain sensitive to order selection and ignore non-Markovian correlations [13, 14] that carry additional predictive signal beyond what network topology alone captures. Deep learning approaches [1, 6, 15] achieve strong performance through long-range temporal modelling, but require full offline training and offer no lightweight mechanism for adapting to an ongoing trajectory in real time. In addition, several research have sought to reconcile population- and individual-level mobility patterns. Hawelka et al. [5] showed that collective models built from large, diverse trajectory corpora substantially outperform individual Markov models for users with short histories, highlighting the value of pooling information across the population when individual data is scarce. Qiao et al. [11] proposed a hybrid variable-order model that adapts to individual behaviour through frequent sequence mining and temporal factors, but relies solely on each user's own historical data without any population-level prior. For transit networks, with IOHMM, Mo et al. [7] capture individual regularities through a hidden state structure, yet without explicitly combining population- and individual-level components.

Next-node prediction algorithms fall into two broad classes [5]: *individual* methods, such

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as single-user Markov models, which suffer from cold-start when history is limited; and *collective* methods, which exploit population-level patterns to mitigate this limitation but often lack an online adaptation mechanism. The proposed Dual Memory Markov Chain (DM-MC) bridges both by casting them as two complementary memories: a population-level Long-Term Memory (LTM), analogous to the expert ensemble [5], encodes shared mobility priors from the full training corpus, while an individual-level Short-Term Memory (STM), acting as the single-user predictor/expert, is built online from the current trajectory alone. The two are combined through the entropy-weighted dual-memory mechanism of IDyOM [9, 8], allowing DM-MC to dynamically shift reliance from population priors to individual patterns as the trajectory grows, effectively countering the cold-start problem without requiring any additional data sources.

2 Dual-Memory Markov Chain Model

The proposed model estimates $P(v_{t+1} \mid v_1, \ldots, v_t)$ for a trajectory $s = (v_1, \ldots, v_t)$ over a vocabulary $\mathcal{V}$ from $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, by combining two complementary components: the LTM encoding population-level patterns from a training corpus, and a STM capturing trajectory-specific regularities inferred online from the current trajectory.

For each order $k \in \{1, \ldots, K\}$, the context is the last k nodes $c^{(k)} = (v_{t-k+1}, \ldots, v_t)$. Each order produces a distinct k-gram conditional.

The LTM stores corpus k-gram counts $C^{(k)}_{c,v}$ (number of times node v followed context c in the training trajectories) and applies Laplace smoothing:

$$P^{(k)}_{\mathrm{LTM}}(v) = \frac{C^{(k)}_{c^{(k)},v} + \varepsilon}{\sum_{v'} \big(C^{(k)}_{c^{(k)},v'} + \varepsilon\big)}. \tag{1}$$

For $k = 1$ this reduces to a row of the smoothed first-order transition matrix $T_{uv} = (C_{uv} + \varepsilon) / \sum_{v'} (C_{uv'} + \varepsilon)$. If the k-gram $c^{(k)}$ is unseen in the corpus, the model backs off to order 1.

The STM scans the current trajectory $v_{1:t}$ for earlier occurrences of the same context $c^{(k)}$ and counts what node followed each match, $n^{(k)}(v) = \sum_{i=1}^{t-k} \mathbf{1}[(v_i, \ldots, v_{i+k-1}) = c^{(k)}]\, \mathbf{1}[v_{i+k} = v]$, normalised with Laplace smoothing:

$$P^{(k)}_{\mathrm{STM}}(v) = \frac{n^{(k)}(v) + \varepsilon/V}{\sum_{v'} \big(n^{(k)}(v') + \varepsilon/V\big)}. \tag{2}$$

If $c^{(k)}$ has no prior occurrence in $v_{1:t}$, $P^{(k)}_{\mathrm{STM}}$ defaults to the uniform distribution over V nodes.

Orders $k = 1, \ldots, K$ and the two memories are aggregated via the same mechanism: each distribution P_m receives a weight α inversely proportional to its Shannon entropy $(H(P_m) = -\sum_v P_m(v) \log P_m(v))$, $\alpha_m = \frac{H(P_m)^{-1}}{\sum_{m'} H(P_{m'})^{-1}}$, so that more confident (lower-entropy) distributions contribute more. Applying this first within each memory across orders yields P_{LTM} and P_{STM}; a second application across the two memories gives the final prediction:

$$P_{\mathrm{DM}}(v) = \alpha_{\mathrm{LTM}}\, P_{\mathrm{LTM}}(v) + \alpha_{\mathrm{STM}}\, P_{\mathrm{STM}}(v). \tag{3}$$

This allows DM-MC to dynamically balance global mobility priors from the population with trajectory-specific patterns as more context accumulates.

3 Methodology

We evaluate DM-MC against AKOM [10], CPT+ [4], MOGen [2], IOHMM [7], and entropy-weighted Markov chains (HW-MC). The latter use maximum orders $k \in 1, 5, 10$ and combine multi-order Markov predictions using entropy-based weights that favor lower-uncertainty contexts. We consider two complementary prediction tasks: *next-node prediction* and *auto-regressive trajectory prediction*. All models are fitted on 90% of the data and evaluated on the remaining 10% under a 5-fold cross-validation setup over each datasets described in Section 3.1. For DM-MC, experiments are conducted using three Markov-chain orders ($k \in \{1, 2, 3\}$) and three node encoding schemes described in Section 3.2.

3.1 Datasets

Experiments are conducted on four cities of varying scale and mobility patterns: New York, Frankfurt, Singapore, and Abu Dhabi. Mobility trajectories are drawn from the WorldMove dataset [16]. For each city, we retrieve the corresponding road network from OpenStreetMap and map each raw trajectory onto it, so that every trip becomes a sequence of OSM node identifiers. Statistics of the resulting datasets are summarised in Table 1.

Table 1 Statistics of the four datasets after map-matching to OpenStreetMap.

City	$\lvert\mathcal{V}\rvert$	$\lvert\mathcal{E}\rvert$	Trips	Users	Days	Avg dist (km)	Avg dur (min)
New York (US)	2,116	23,736	165,328	500	15	13.71	27.42
Frankfurt (DE)	1,479	16,500	39,621	500	15	12.62	25.25
Singapore (SG)	1,802	20,452	90,355	500	15	14.40	28.80
Abu Dhabi (AE)	2,565	28,770	21,759	500	15	14.54	29.08

3.2 Node Encoding

We consider three encoding schemes. In the *categorical ID encoding*, each OSM node is assigned a unique integer identifier $v_t \in \{1, \ldots, N\}$ with $N = |\mathcal{V}|$, and a k-th order Markov chain models $P(v_{t+1} \mid v_{t-k+1}, \ldots, v_t)$. In the *higher-order (de Bruijn) encoding*, the vocabulary is expanded to k-grams of consecutive nodes, where an order-k state is represented as $\tilde{v} = (v_{t-k+1}, \ldots, v_t) \in \mathcal{V}^k$, and treated as a single categorical token, enabling longer spatial dependencies to be encoded directly in the state space. In the *graph-embedding encoding*, each node is represented by a Node2Vec [3] embedding augmented with its discretized bearing angle, $\phi(v_t) = [\mathbf{e}_t \,\|\, \theta_t] \in \mathbb{R}^{d+1}$, where $\mathbf{e}_t$ is the Node2Vec embedding of node v_t and θ_t its bearing angle.

3.3 Next-Node Prediction

The first task evaluates local prediction accuracy. Given a context window of the previous 10 nodes: $X_t = (v_{t-9}, \ldots, v_t)$, the model predicts the immediately following node v_{t+1}. All possible sliding windows are extracted from each test trajectory, generating a set of context–target pairs. Performance is measured using *Accuracy@k*, and *Mean Reciprocal Rank (MRR)*. *Accuracy@k* corresponds to the proportion of samples for which the true next node appears within the top-k ranked predictions. *MRR* is defined as: $\mathrm{MRR} = \frac{1}{N}\sum_{i=1}^{N} \frac{1}{\mathrm{rank}(y_i)}$, where $\mathrm{rank}(y_i)$ denotes the position of the true next node in the ranked prediction list.

3.4 Auto-Regressive Trajectory Prediction

The second task evaluates long-horizon forecasting, the first 10 nodes of a trajectory are provided as an initial seed. At each step, the model predicts the next node and feeds its own prediction back into the context: $X_{t+1} = (v_{t-8}, \ldots, v_t, \hat{v}_{t+1})$, where $\hat{v}_{t+1}$ denotes the predicted node. Because future predictions depend on previous model outputs rather than ground-truth observations, errors may propagate over time, making this task more challenging than next-node prediction. Performance is measured using *AR@k* and *AR-MRR*. *AR@k* is similar to *Accuracy@k*, while *AR-MRR* follows the same formulation as *MRR*, but both are averaged over T, the total number of auto-regressive prediction steps.

4 Results

As reported in Table 2, Figure 1, and Figure 2, DM-MC consistently matches or outperforms the considered baselines across both experimental settings. In the next-node prediction task, HW-MC exhibit a gradual degradation from *Acc@1* $= 0.6646$ (MC1) to 0.6228 (MC10), reflecting the increasing sparsity of higher-order k-grams over a large vocabulary. In contrast, DM-MC with Categorical-ID encoding remains remarkably stable around 0.654 across all orders, showing that its dual-memory back-off mechanism effectively incorporates higher-order information without suffering from severe sparsity effects. The negligible differences between orders $k = 2$, 5, and 10 further suggest that a low-order configuration is sufficient for road-network prediction. DM-MC$_2$ with graph embedding achieves the best overall performance (*Acc@1* $= 0.6884$, *MRR* $= 0.7670$), indicating that combining the dual-memory formulation with a topology-aware representation provides the largest performance gains.

Table 2 Prediction results averaged over 4 datasets. Left: next-node. Right: auto-regressive.

Model	Encoding	Acc@1	Acc@3	Acc@5	MRR	AR@1	AR@3	AR@5	AR-MRR
DM-MC2	Cat. ID	0.6549	0.8089	0.9394	0.7503	0.2857	0.3455	0.4070	0.3285
DM-MC2	De Bruijn	0.6572	0.7998	0.8874	0.7416	0.2500	0.2676	0.2809	0.2609
DM-MC2	Graph Emb.	**0.6884**	0.8048	0.8614	**0.7670**	**0.3405**	0.4751	0.5702	**0.4559**
DM-MC5	Cat. ID	0.6536	0.8088	0.9394	0.7497	0.2859	0.3454	0.4071	0.3288
DM-MC5	De Bruijn	0.6557	0.7998	0.8874	0.7408	0.2500	0.2676	0.2809	0.2609
DM-MC5	Graph Emb.	0.6858	0.8048	0.8614	0.7657	0.3400	0.4761	0.5699	0.4556
DM-MC10	Cat. ID	0.6533	0.8087	0.9394	0.7496	0.2859	0.3454	0.4071	0.3288
DM-MC10	De Bruijn	0.6557	0.7998	0.8874	0.7408	0.2500	0.2676	0.2809	0.2609
DM-MC10	Graph Emb.	0.6858	0.8041	0.8613	0.7654	0.3400	0.4760	0.5701	0.4556
AKOM	Cat. ID	0.6535	0.7997	0.9298	0.7453	0.2830	0.3461	0.4075	0.3265
CPT	Cat. ID	0.6396	0.8003	0.9323	0.7385	0.2809	0.3439	0.4042	0.3245
IOHMM	Cat. ID	0.6333	0.7925	0.9029	0.7317	0.2793	0.3362	0.3843	0.3226
MOGEN	Cat. ID	0.6520	0.7999	0.9315	0.7447	0.2845	0.3466	0.4059	0.3273
HW-MC1	Cat. ID	0.6646	0.8121	0.9567	0.7592	0.2881	0.3532	0.4143	0.3342
HW-MC5	Cat. ID	0.6469	0.8082	0.9496	0.7488	0.2860	0.3475	0.4112	0.3308
HW-MC10	Cat. ID	0.6228	0.8074	0.9482	0.7363	0.2814	0.3429	0.4047	0.3262

In the auto-regressive setting, all categorical models converge toward a shared performance ceiling of *AR@1* ≈ 0.28, reflecting the average out-degree of road intersections compounded by error accumulation. DM-MC2 with graph embedding is the only configuration that breaks through this barrier, reaching *AR@1* $= 0.3405$ and *AR-MRR* $= 0.4559$, a $\sim$20% absolute gain over all other configurations, by compressing the node vocabulary into region-level clusters and thereby reducing the effective branching factor. This highlights that the performance advantage stems from the synergy between the dual-memory predictor and a topology-aware encoding. In contrast, De Bruijn encoding degrades sharply under auto-regressive conditions due to the error propagation over its large edge vocabulary. Overall, these preliminary results suggest that DM-MC outperforms existing methods across both prediction scenarios.

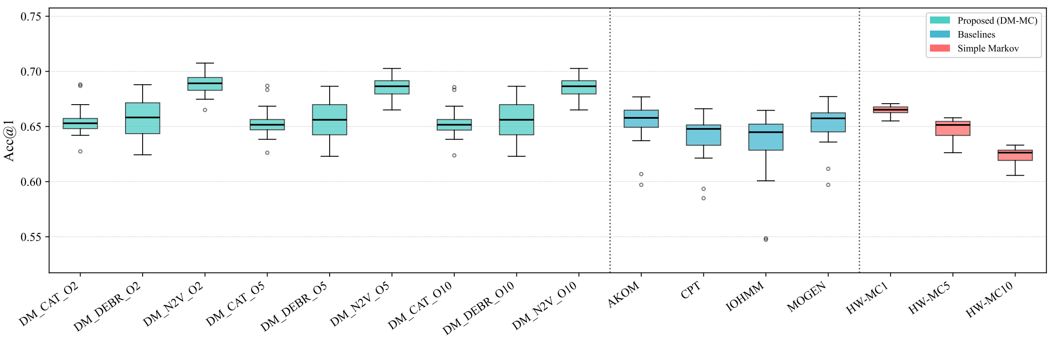

Figure 1 Next-node prediction scores *Accuracy*@1 averaged over the 4 datasets.

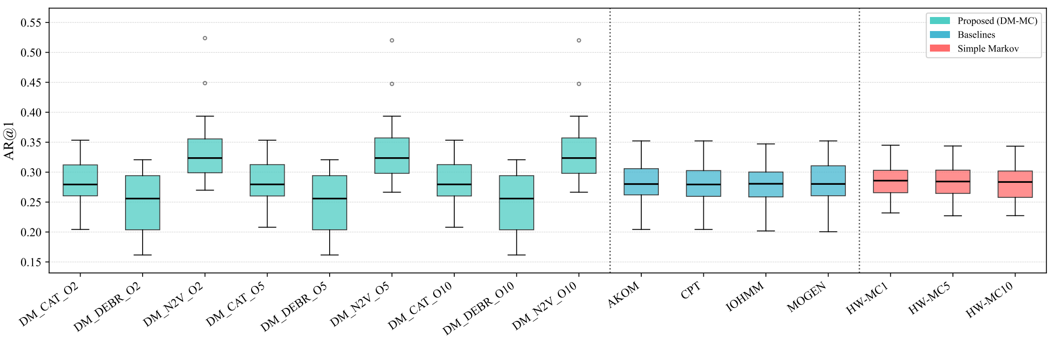

Figure 2 Auto-regressive prediction scores *AR*@1 averaged over the 4 datasets.

5 Conclusion & Future Work

We present Dual-Memory Markov Chain (DM-MC) model that combines long-term population-level transitions with short-term trajectory-specific statistics via entropy-weighted aggregation over multiple orders. In prelimary results, DM-MC improves over HW-MC and baselines on both next-node and auto-regressive prediction tasks, with stronger gains under long-horizon forecasting, indicating improved robustness to error accumulation. Future work will focus on adding temporal decay in the STM, learning user- or cluster-specific LTMs to capture heterogeneity/groups, evaluating the sensitivity of the model to varying context window lengths, and conducting a more detailed LTM/STM contribution analysis (e.g., via regret-based measures [5]).

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